



RFP: Castle Rysen Coffee

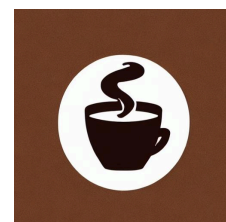
Design, Implementation, and Support of a New Network Infrastructure
for the Last Remaining Coffee Shop of the Future

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Castle Rysen Coffee

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Overview

Nestled amidst the ruins of a dystopian world, Castle Rysen Coffee stands as the last bastion of hope. Having discovered the secret to resurrecting the once-extinct coffee bean, Castle Rysen Coffee has not only revitalized the cherished elixir of life but has also imbued it with properties that shield weary humans from the radiation poisoning that plagues the desolate metropolitan landscapes. As the sole provider of this life-sustaining brew, Castle Rysen Coffee looks to maintain and expand its vital role in this harsh new world. Castle Rysen Coffee is now seeking proposals from proficient vendors to create, implement, and support a state-of-the-art network infrastructure that will solidify its command over the caffeinated realm.

The objective of this Request for Proposal (RFP) is to summon proposals from proficient vendors for the concoction, implementation, and sustenance of an innovative network infrastructure for Castle Rysen Coffee's central command bunker, two auxiliary fallout shelters which provide oversight for the coffee outposts (each fallout shelter sustains between three to ten individual coffee outposts within the desolate wasteland). The revamped network infrastructure must be impenetrable, expandable, and equipped to accommodate present and future business needs in the bleak coffee-scarce world.

As the sole provider of caffeine in the new world, Castle Rysen Coffee is experiencing rapid growth and plans to expand its operations. The current network infrastructure is lacking, and the organization requires a new network infrastructure that can support its command shelter, two fallout shelters (which provide resources for the district shops), and the individual district shops.

Goals

The primary objectives of this project are to:

1. Concoct and execute an adaptable and impervious network infrastructure that fulfills the demands of each Castle Rysen regional establishment:
 - a. Castle Rysen Central Office (~**200 Humans**)
 - b. Castle Rysen Fallout Shelter (~**50 Humans**)
 - c. Castle Rysen District Shop (~**15 Humans**)

Each Central Office establishes core services for a region and is able to provide support for **30 Fallout Shelters**. Each Fallout Shelter is able to provide support for **50 District Shops**.

2. Provide all necessary documentation for the project, including but not limited to network diagrams, IP addressing spreadsheets, wireless coverage maps, and device credentials.
3. Create hardware and configuration standards for all network infrastructure devices (switches, routers, WAPs, etc.) allowing turnkey rollouts with minimal error.
4. Establish unbroken connectivity amongst central command, auxiliary fallout shelters, and coffee shops. Internet connectivity should be resilient in all areas that still have the capability of wired and wireless / cellular Internet connectivity.
5. Each District Shop network should support local Plex-based video streaming service for patrons AND video surveillance of every coffee house. Between 3 - 5 cameras will be connected at each coffee house, depending on the size.

6. Employ fitting security measures to safeguard the network against both internal insurrection and external marauders.
 - a. Each **District Shop** should have distinct network segments for internal communication and guest access. Guest access should be restricted to Internet and Plex server access.
 - b. Each **Fallout Shelter** should have network segments for network management traffic, internal communication, video surveillance, and guest access. Guest access should be restricted to Internet access only. The network management segment should be self-contained and be the only one able to administratively access network devices.
7. Optimize network loop prevention and performance for a plethora of dystopian applications and services, ensuring the smooth running of Castle Rysen Coffee's operations in the bleak, caffeine-craving world.
8. Vigilant monitoring of essential elements of each coffeehouse network to track trends and ensure local and Internet connectivity is maintained.

Specifications

The scope of work for this project includes the following tasks:

- Assessment of current network infrastructure and identification of required improvements.
- Design of a new network topology, including routers, switches, firewalls, access points, and controllers.
- Implementation of IPv4 and IPv6 addressing, routing protocols, and VLANs.
- Configuration of interswitch connectivity and Layer 2/Layer 3 EtherChannels.
- Installation and configuration of wireless infrastructure, including access points and controllers.
- Integration of network services, such as STP, NAT, DHCP, DNS, and NTP.
- Implementation of network security measures, including access control lists and Layer 2 security features.
- Utilization of automation and programmability to deploy networks when possible.
- Supply of ongoing monitoring, support, and maintenance services for the novel network infrastructure, ensuring Castle Rysen Coffee's continued dominance in the dystopian, radiation-riddled world.

During the unfolding of this venture, Castle Rysen Coffee may unveil Alteration Petitions or Service Inquiries grounded on newfound necessities or barricades encountered in the dystopian terrain. The contender ought to be adept at responding and assimilating these entreaties into the project with haste, ensuring the relentless progress of Castle Rysen Coffee's network infrastructure endeavors.

Milestones

The project should be divided into the following phases.

Phase 1: Project Requirements and Network Design

1. Gather requirements: Understand the organization's network needs, such as the number of users, departments, applications, security, and performance requirements.
2. Design the network topology: Based on the requirements, design a network topology that includes the appropriate architecture (two-tier, three-tier, or spine-leaf) and incorporates routers, switches, firewalls, access points, and controllers.
3. Design scalable IP addressing for current IPv4 environment with recommendations for future IPv6 compatibility.

Phase 2: Network Implementation

1. Connect and configure network devices: Set up routers, switches, WAPs, and other devices with appropriate configurations, including IPv4 and IPv6 addressing, VLANs, trunk ports, and EtherChannels.
2. Initially provide Internet access using a single, public IPv4 address the district coffee shop. In larger regions, the district shops may receive Internet access through the uplink fallout shelter.
3. Implement routing protocols: Configure static routing and OSPFv2 for IPv4 and IPv6 to ensure effective and efficient routing in the network.
4. Set up wireless infrastructure: Install and configure wireless access points, controllers, and WLAN components for the organization.

Phase 3: Network Services and Security

1. Configure IP services: Set up NAT, DHCP, DNS, NTP, and remote access (SSH) for each district shop.
2. In each district shop, the cafe patron devices should be separated from the administrative devices (such as network equipment, cameras, Plex server, etc.) with a security boundary. Patrons should be allowed to reach the Plex server, but only using the specific ports needed to view the catalog and stream video (TCP: 443, 32400, 32469; UDP: 1900, 5353).
3. Ensure SSH and Telnet access to all network devices is limited to only devices on the Admin VLAN (Cafe) and Management VLAN (Fallout).
4. Implement network security: Enable core Layer 2 security features such as port security, DHCP Snooping, and Dynamic ARP Inspection (DAI).
5. Set up wireless security protocols and VPNs for secure remote access.
6. Implement network monitoring: Configure SNMP, syslog, and QoS features to monitor and manage the network effectively.

Phase 4: Network Automation and Programmability

1. Introduce automation: Learn and implement automation techniques using configuration management tools like Puppet, Chef, or Ansible.
2. Explore controller-based networking: Understand the differences between traditional and controller-based networking and experiment with software-defined architectures.
3. Work with REST-based APIs: Explore how to use REST-based APIs to manage network devices and learn the characteristics of JSON-encoded data.

Phase 5: Network Troubleshooting and Optimization

1. Troubleshoot network issues: Identify and resolve common issues related to interfaces, cables, routing, and device configurations.
2. Optimize network performance: Analyze network performance and make necessary adjustments to routing, switching, and security configurations to ensure optimal performance.

Phase 6: Project Documentation and Closeout

1. Document the network: Create comprehensive network documentation, including topology diagrams, device configurations, and a summary of implemented services and security measures.
2. Closeout and review: Perform a final review of the project, discuss lessons learned, and identify areas for improvement or further learning.

Castle Rysen Coffee - Floorplan

All Castle Rysen Coffee house locations are built to the same specification. The following floorplan is used for all newly constructed locations. Occasional retrofit buildings are used in locations where building materials are not available.



The walkthrough of a Castle Rysen cafe is an elegant experience. Shelves are filled with the remaining books extracted from the piles of rubble that once were thriving libraries. The scent of the long-lost Arabica beans wafts through the air, mingling with the gentle aroma of old parchment. The furniture, crafted from salvaged wood, tells its own story of resilience, each piece bearing the etchings of a past era. Amid the remnants of the past, the heart of the coffee house lies in its state-of-the-art network center, cleverly hidden behind a well-preserved mural.

Castle Rysen Fallout Shelter - Floorplan

All Castle Rysen Fallout Shelters are built to the same specification. The following floorplan is used for all newly constructed locations. Occasional retrofit buildings are used in locations where building materials are not available.



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